

DATUM LINE

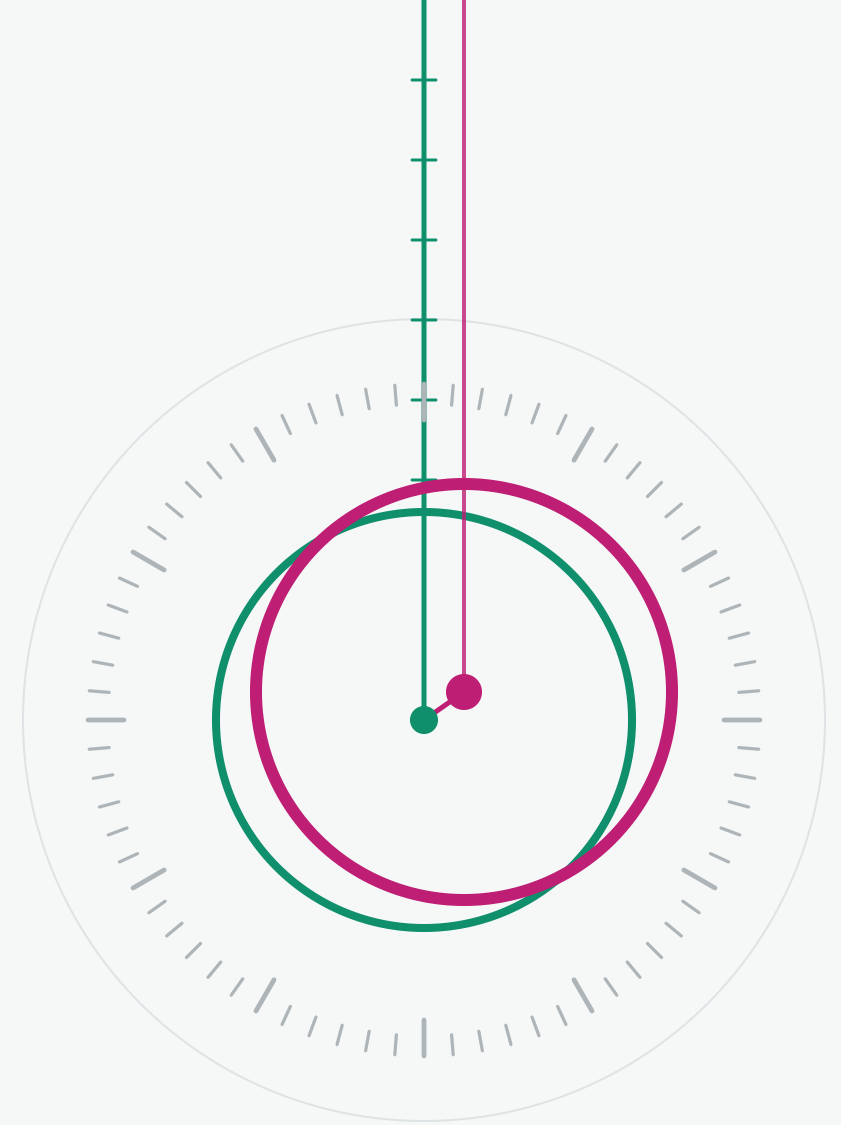
AI × BIO · 29 SEPT

Control is an illusion.

Confidence without control.

● your cells ○ the field

Kat Titterton · San Francisco





macrofluidic.me/resume

Katherine Titterton

Five labs, and the same **unanswered question** at every bench. It made the science slow and wrong.

- 2023–2026 **Axiom** SAN FRANCISCO
founding MTS • hepatocytes at scale
- 2022–2023 **insitro** SOUTH SAN FRANCISCO
neuroscience assays • productionization
- 2021–2022 **MIT** SYNTHETIC NEUROBIOLOGY
expansion microscopy
- 2018–2021 **abbvie** NEUROSCIENCE
tau • neurons
- 2017–2018  HARVARD MEDICAL SCHOOL
Xenopus oocytes • GABA_A receptors
- 2017 **COLGATE** BA
neuroscience + math biology

MY WORK

- 1 Cell Painting in primary human hepatocytes
Cell Systems 2026 • Axiom Bio
- 2 Counting cells predicts bioactivity benchmarks
Nature Communications 2026 • Axiom Bio
- 3 A neuronal tau aggregation assay
J Biol Chem 2025 • AbbVie
- 4 Multiplexed expansion revealing
Nature Communications 2024 • MIT
- 5 Neurosteroid photolabeling of GABA_A receptors
Eur J Med Chem 2019 • MGH • Harvard Medical School

PUBLISHED WITH

Broad Institute • MIT • AbbVie • EMBL-EBI • HESI • Bayer
• University of Cambridge • Uppsala University

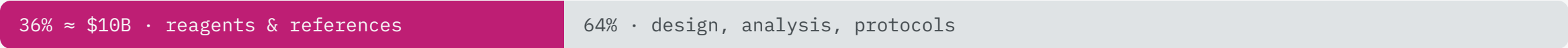
“Do my cells look normal?”

WHY NOW

Since 2023 the field’s reference is public [7].

IRREPRODUCIBLE PRECLINICAL RESEARCH · US · PER YEAR [6]

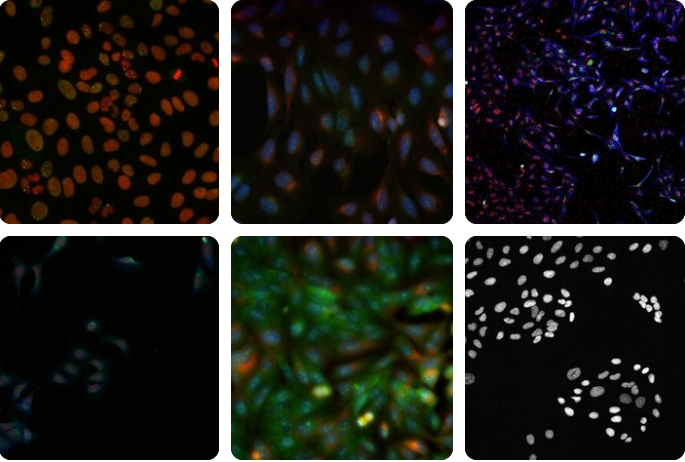
\$28B



1

search

Q do my U2OS cells look normal



2

ask around

To: whoever ran this line last

Subject: does this look right to you?

Attached: screenshot.png

3

guess

\$ RUN IT ANYWAY · TOO FAST

hundreds of thousands

of R&D before the SOP was fixed

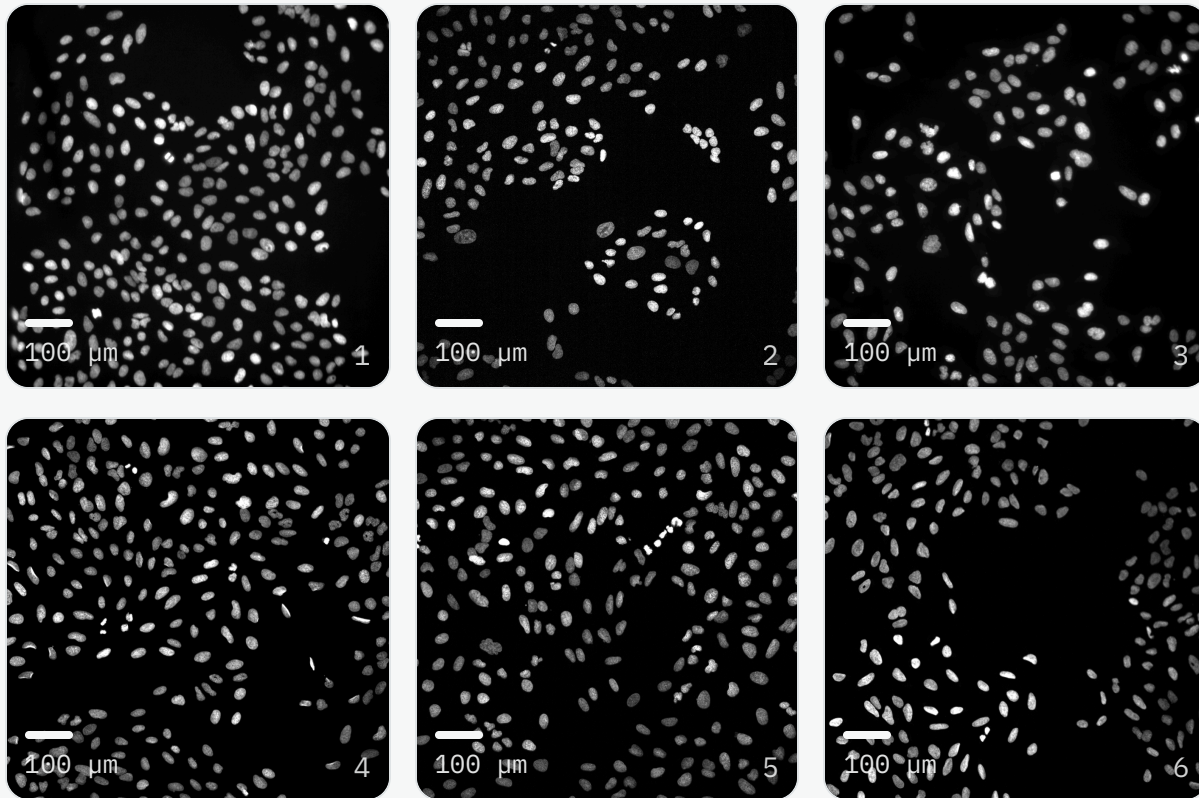
🕒 HOLD · TOO SLOW

months

developing cells that were already good enough

[6] Freedman et al., *PLoS Biol*, 2015. Search results: real U2OS images from six papers.

One lab, or six?



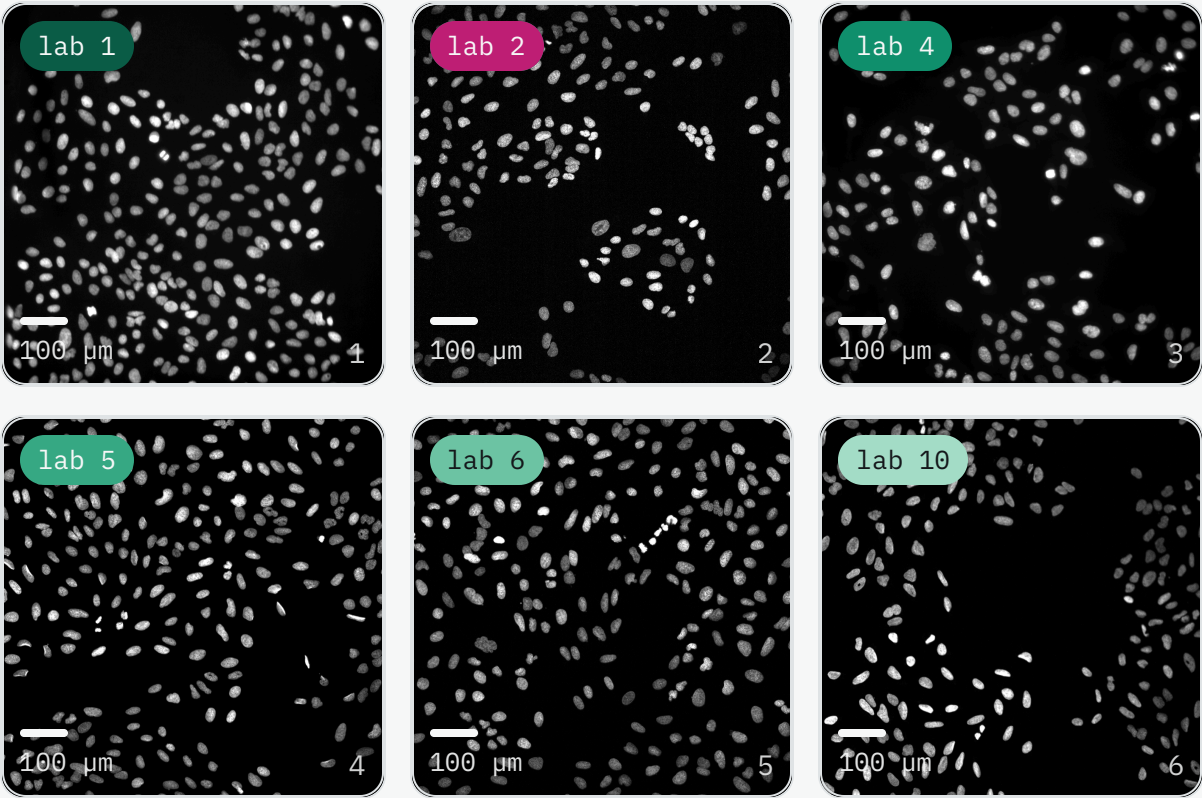
Cell Painting five dyes, 3,773 numbers per well

-  same line ✓
-  same protocol ✓
-  no drug ✓

Fig. 1 93,228 untreated wells, 11 labs, plates held out. JUMP [7]

[click → the answer](#)

Could you tell? A classifier can.



Cell Painting five dyes, 3,773 numbers per well

99.9% CLASSIFIER

9% CHANCE, 1 IN 11 LABS

78 wells wrong, of 93,228

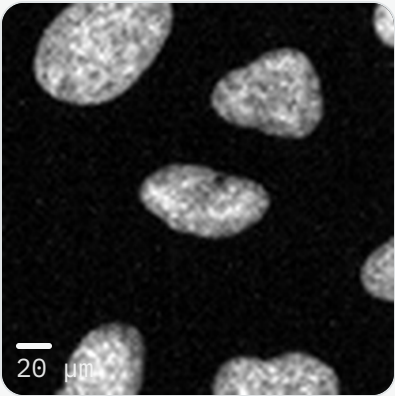
Normalise each plate to its own controls: 99.9% becomes 24%. A3

Fig. 1 93,228 untreated wells, 11 labs, plates held out. JUMP [7]

click → follow one well

One well, **all the way out.**

ONE NUCLEUS



192 μm^2 · 0.59

ITS WELL

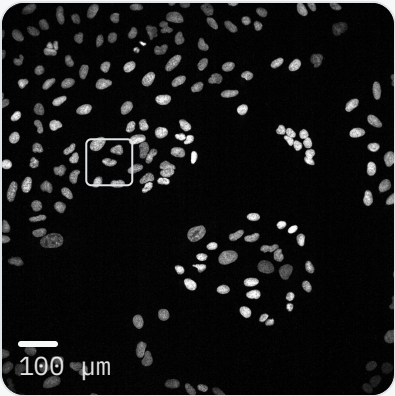
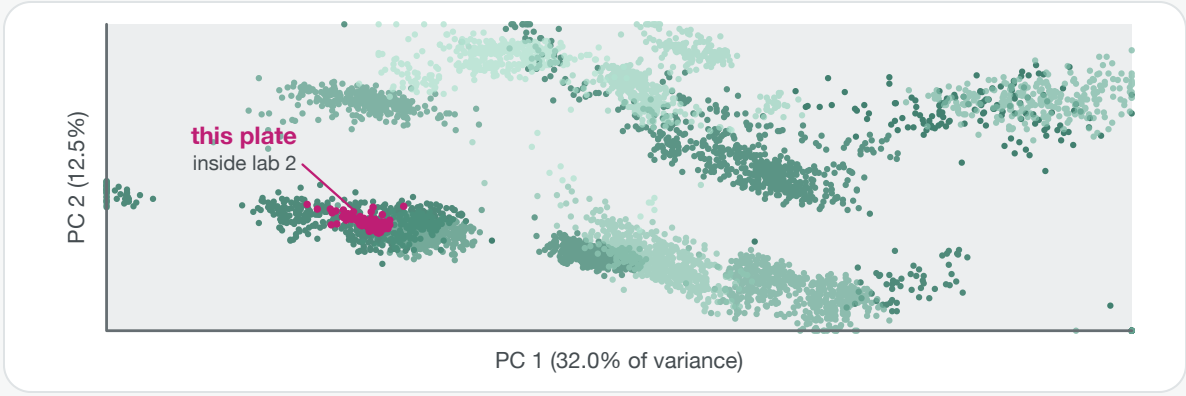


plate 1053600681

3,773
numbers
→
one point

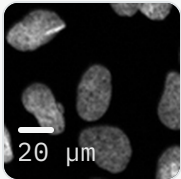
WHERE IT LANDS



93,228 wells, 11 labs

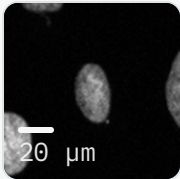
other labs

lab 5



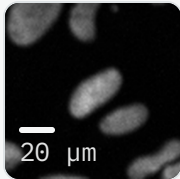
195 μm^2 · 0.94

lab 6



213 μm^2 · 0.50

lab 10



207 μm^2 · 0.29

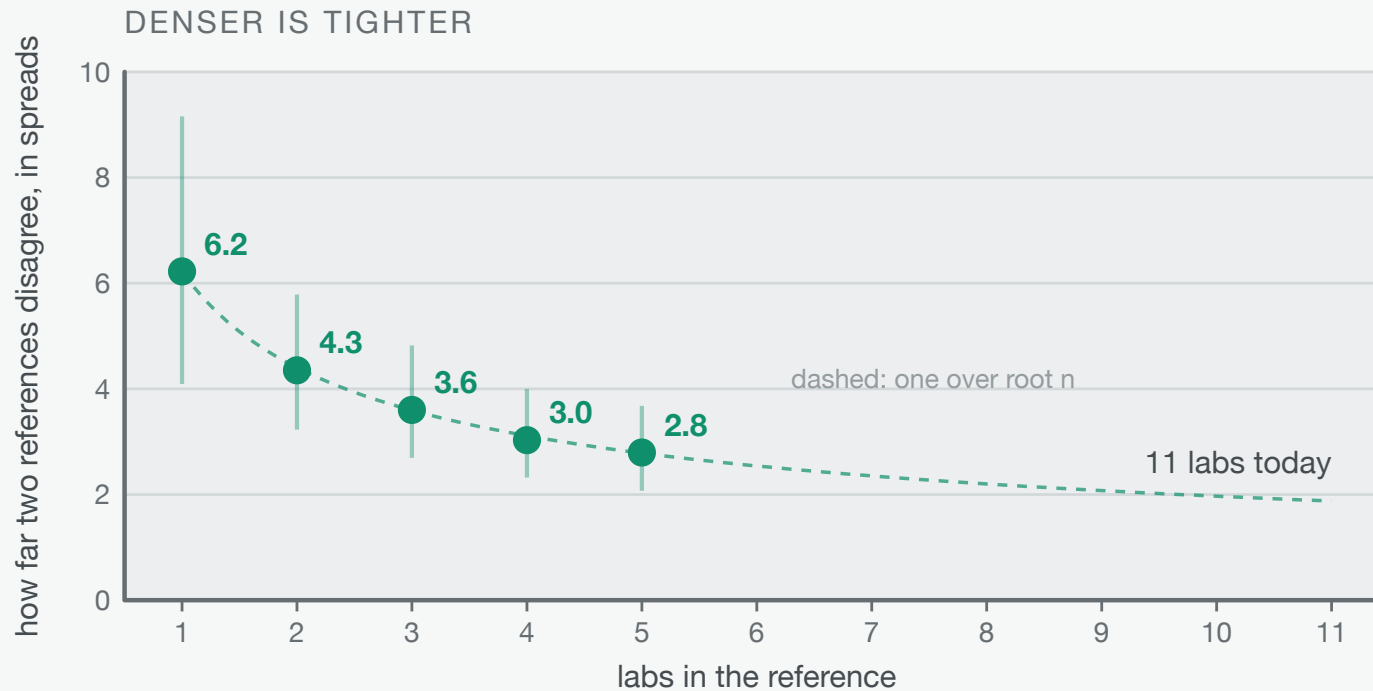
AREA $\approx \approx$
TEXTURE $\neq \neq$

One lab's normal is not another's. The difference is shape, not count.

Fig. 1 93,228 untreated wells, 11 labs, plates held out. JUMP [7]

click → start over

Put them in one frame. **It tightens.**



THE REFERENCE

6.2 → 2.8

from one lab to five:
every lab that joins tightens
the baseline for everyone

your own baseline

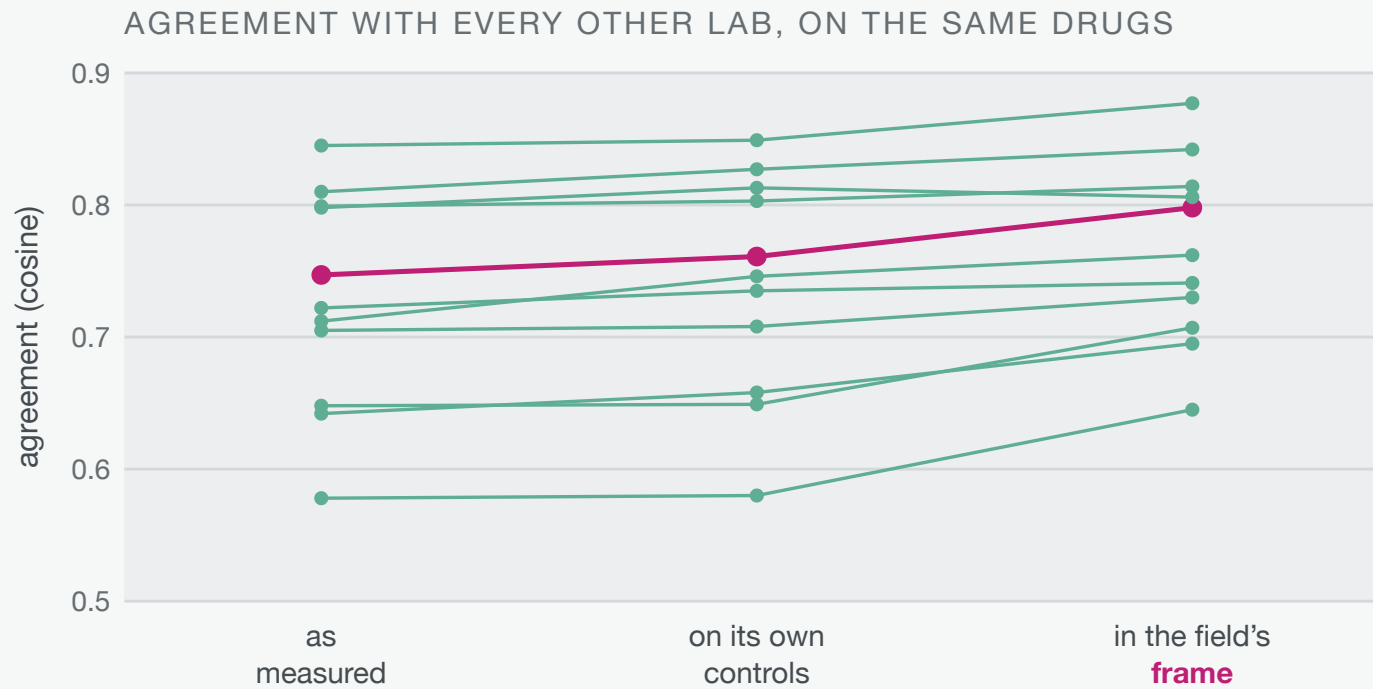
0.9 → 0.4

from one plate to eight, then flat

Fig. 3 Two references, each built from the same number of labs. 1,871 plates, 11 labs. JUMP [7]. More: A14

click → the answers

In one frame, answers move closer.



EVERY LAB MOVES UP

0.75 → 0.78

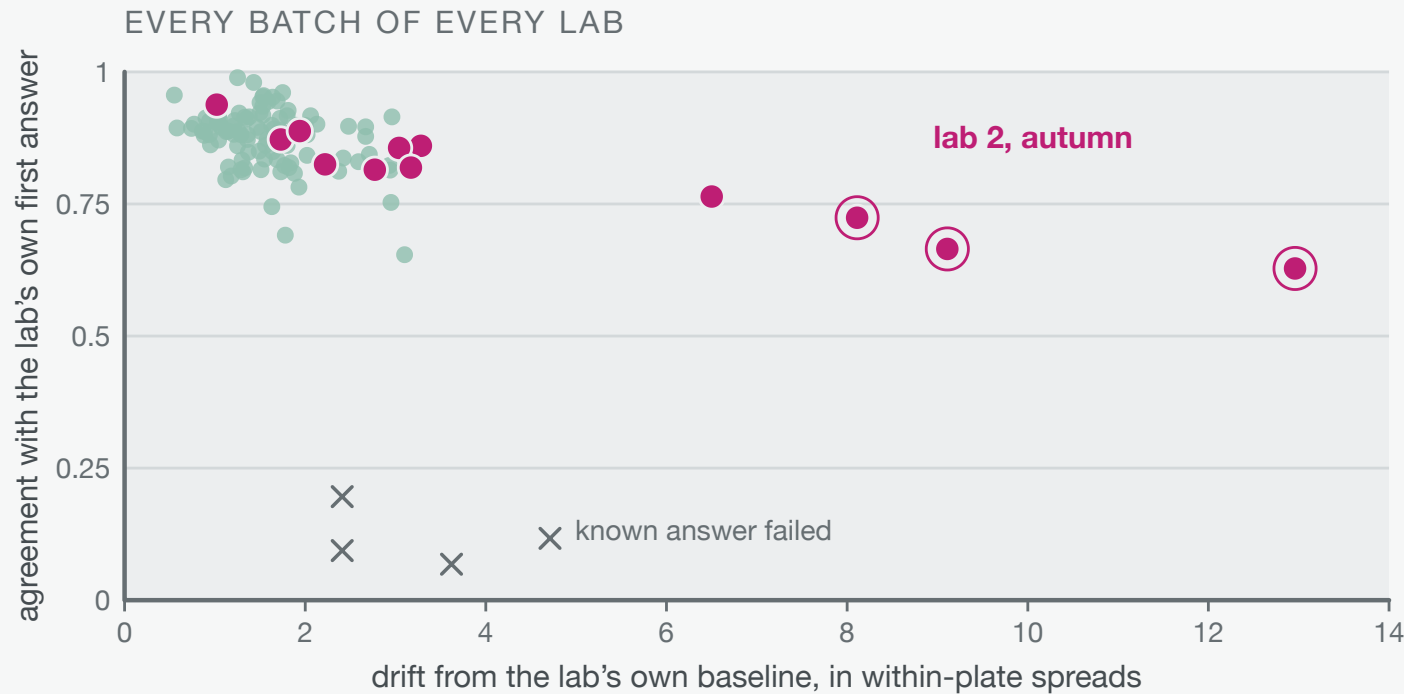
same drugs, closer answers:
higher in 121 of 125 batches
11 lines, one per lab

against its own controls

0.76 → 0.78

higher in 107 of 125 batches

Where you sit predicts what you measure.



INSIDE LABS

$$\rho = -0.71$$

the further from your baseline,
the further from your own answers

lab 2 against its own June answer

0.85 before the flag

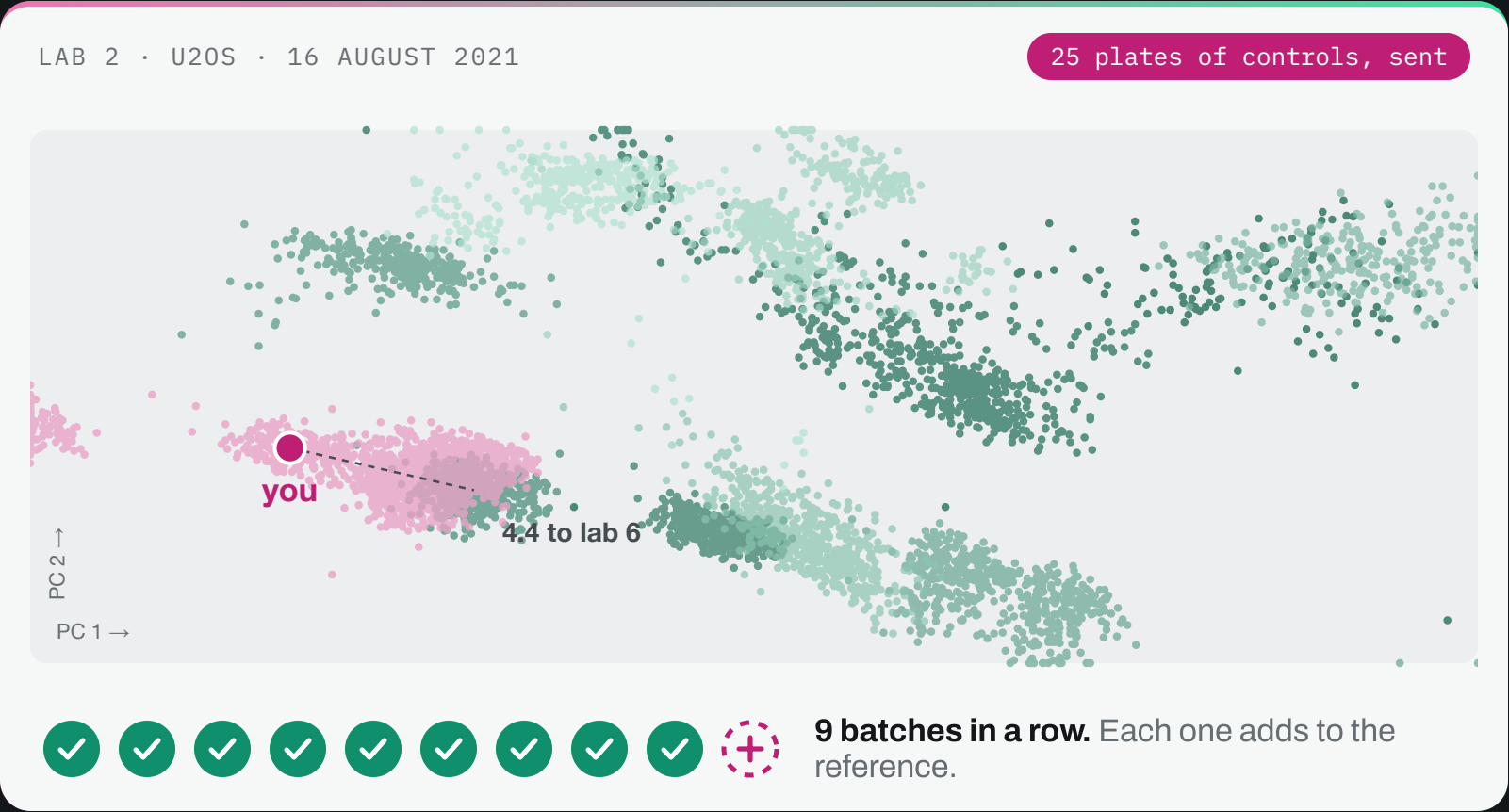
0.67 after it

Fig. 5 One point per batch: 118 batches, 11 labs. JUMP [7]. More: A8, A13

click → start over

Send your controls. See where you stand.

The images you already take. Back: your place in the field, and your data in its frame.



Certificate

LAB 2 · 16 AUGUST 2021



YOUR CELLS
U2OS · 25 plates

WHERE YOU SIT
in distribution
2.8 from your baseline · 4.4 to the nearest lab

YOUR OWN CHECK
188 cells per well · in range

YOUR REFERENCE
1,652 plates · 10 labs

YOUR DATA, IN THE FIELD'S FRAME
agreement with other labs 0.70 → 0.77

Held by nobody who sells you cells,
media or the instrument.

Fig. 6 Lab 2 on 16 Aug, replayed against the published field. JUMP [7]. More: A11

Identity, yes. Behaviour, no.

	STR authentication	Vendor certificate	Plate's own controls	Phenomics platforms	Datum Line
Which line it is	✓	✓	—	—	✓
How it behaves, the whole cell	—	—	✓	✓	✓
Against every lab running the same line	—	—	—	✗	✓
From images you already take	—	—	✓	—	✓
Your data, in the field's frame	—	—	—	—	✓
A baseline that tightens with every lab	—	—	—	—	✓

✓ yes ✓ with a partner: run STR ✗ removed by design: these platforms correct lab differences away

Three buyers, one certificate.

Contributors use it free. The labs that cannot afford to be wrong pay.

	 ACADEMIC LABS AND CORES	 PHARMA AND CRO SCREENING	 CELL BANKS AND VENDORS
What is at stake	the paper	the screen · ~\$2.6M	the lot
What they pay	free, for contributing controls	~\$5–15k per line, per quarter 170× less than one unchecked screen	a fee per certified lot
What it builds	the reference	the revenue	the network

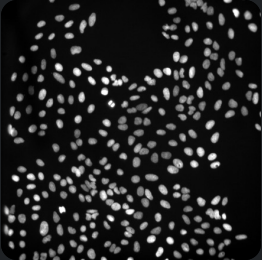
Proposed, not yet validated.

Build the ugliest dataset in biology, **on purpose.**

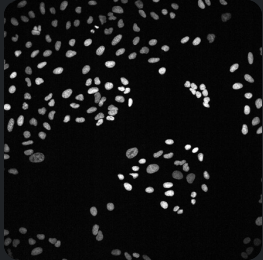
Same cells, same protocol, no drug. It already looks like this.

THE TEST SET · WHAT SIX LABS ALREADY SEND

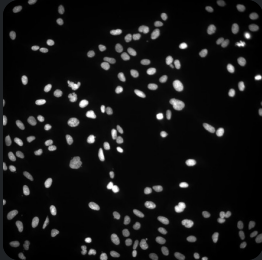
real wells, JUMP cpg0016, CC0



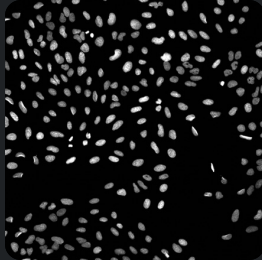
LAB 1
Phenix · widefield



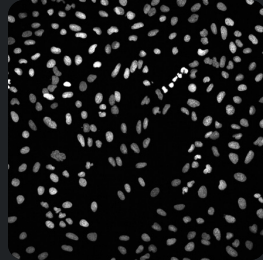
LAB 2
CV8000 · confocal



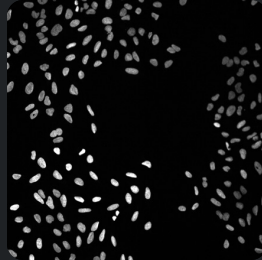
LAB 4
Phenix · widefield



LAB 5
CV8000 · confocal



LAB 6
CV8000 · confocal



LAB 10
CV8000 · confocal

5
microscope models

0.4–1.0
objective NA

0–3
brightfield planes

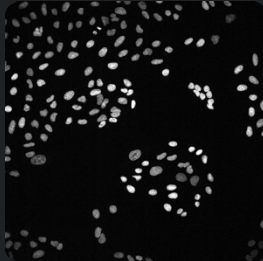
0.60–0.70
µm per pixel

8
plate types

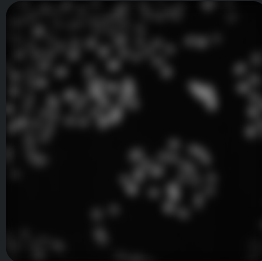
Across the 13 JUMP labs.

RED TEAM

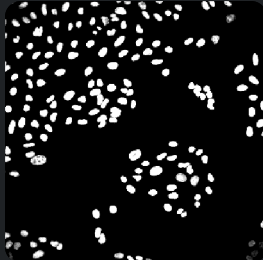
one well, simulated



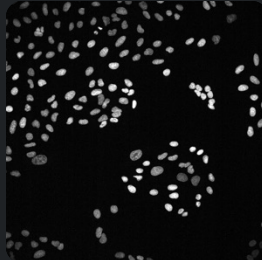
AS IMAGED



DEFOCUSED



OVER-EXPOSED



CLIPPED WELL

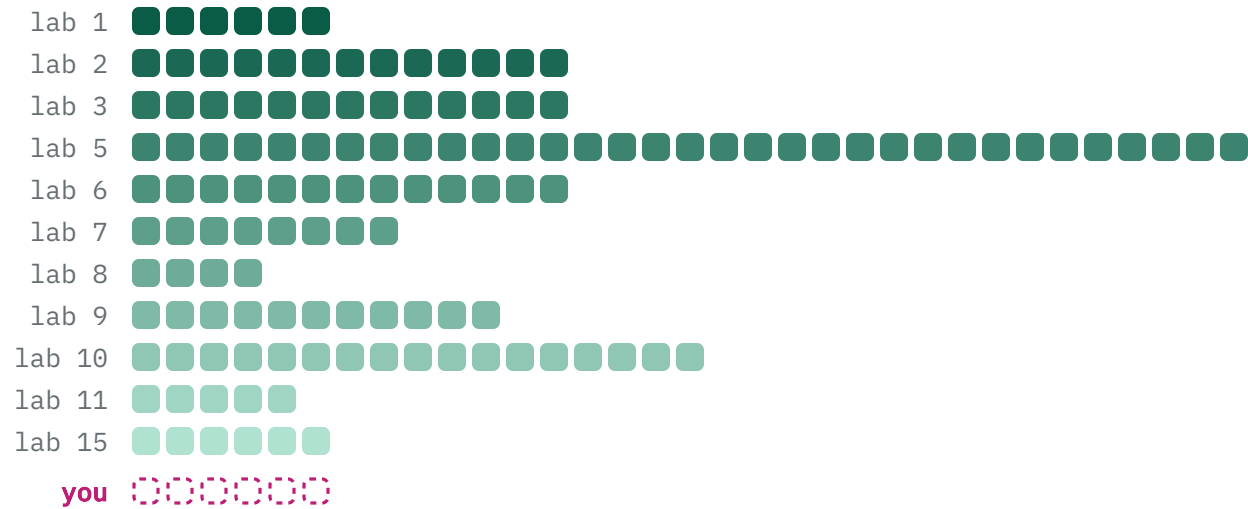
A public benchmark: name the failure.

One shared space for the field, instead of Google Images.

Every plate you image, placed in the field.

A growth chart for cell lines. The reference is what nobody can build alone.

FIG. 7 • ALREADY IN THE FRAME 129 batches • 11 labs • JUMP [7]



1,871 plates, one frame. Every lab that joins tightens it: 6.2 → 2.8.

- NOW
U2OS, fluorescence
- NEXT
more lines, brightfield
- THEN
the baseline a field shares

WHAT I NEED

-  Images U2OS controls
-  People ML, normalization
-  Bench 3-lab proof, then the set
-  Pre-seed to run it

Have images of U2OS cells? Talk to me.

The rest, if you ask.

Find the question, jump to the board. Every number traces to a file in results/ or a numbered reference; pending work says pending.

A1 **Why not trust the plate’s controls?**
There is no control well

A2 **Show me the math.**
Six operations on one embedding

A3 **Why not batch-correct it away?**
What JUMP records

A4 **Is the model the moat?**
Counting cells often wins

A5 **Why not just count cells?**
Two ways to ask one question

A6 **How often do I re-certify?**
The line is not expired

A7 **Isn’t this STR?**
STR says who, images say how

A8 **Is the assay itself working?**
A known answer

A9 **Does it change the result?**
Same drug, different answer

A10 **Does it work without stains?**
Bet on brightfield

A11 **Would it cry wolf?**
Six of 129 batches

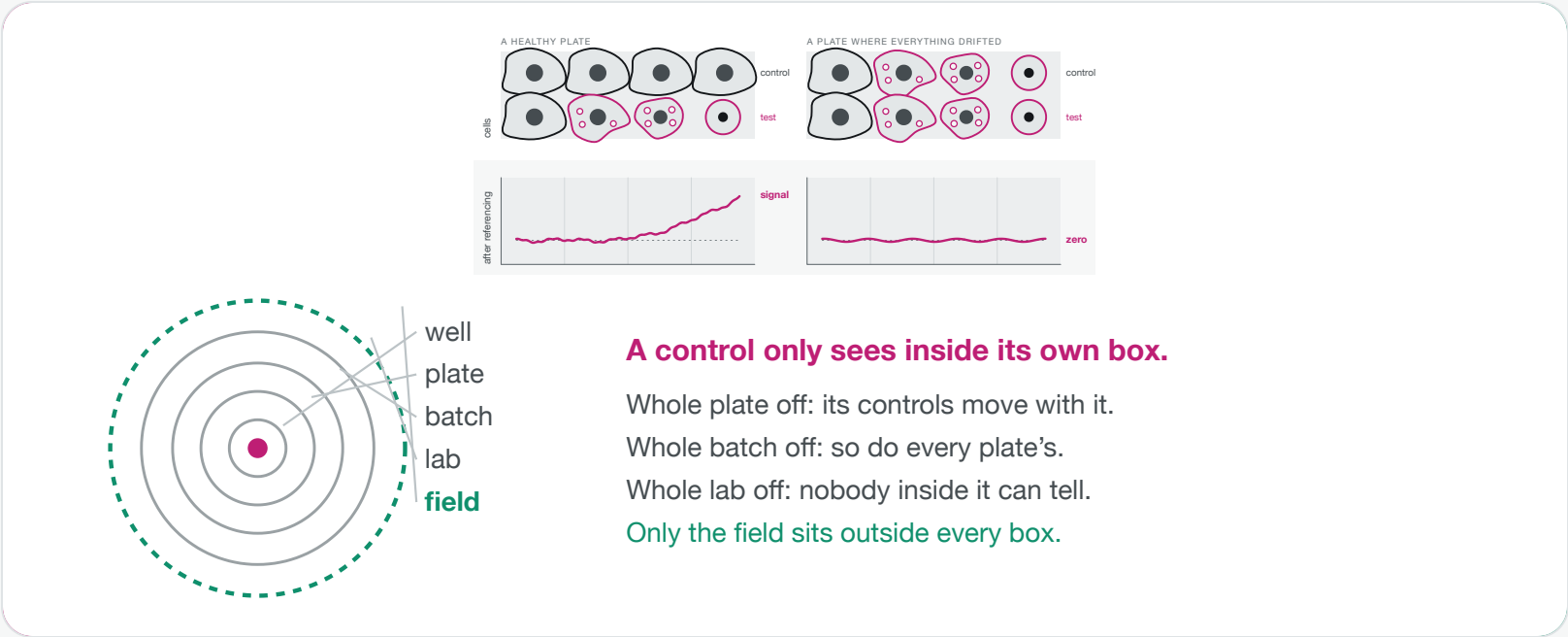
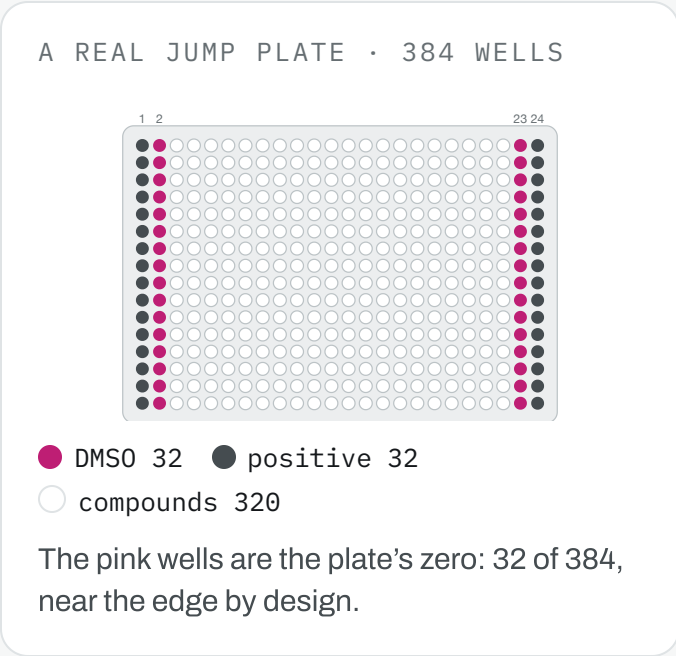
A12 **What moved?**
Stain intensity, not cell count

A13 **How does one lab move over time?**
The lab’s check and the field’s view

A14 **What does the field’s frame buy?**
Agreement between labs

There is no control well.

Every result is test minus control, on the same plate. But a whole plate can be off, or a whole batch, or a whole lab. A control only sees inside its own box.



$$x_{test} = \mu + s + \tau(s) \quad x_{ctrl} = \mu + s \quad \Delta = \tau(s)$$

s , the plate's state, cancels from the number. Never from the cells.

So a control only sees inside its own box. If the plate is sick, so is the control. Then how would anyone know?

Fig. A1 Where a control sits, and what it can see. Left, a real 384-well JUMP compound plate (lab 2, 1053601756): 32 DMSO, 32 positive, 320 compound wells. Right, schematic cells and traces, and the nesting of well, plate, batch, lab and field. No measurements shown.

Six operations on one embedding.

Every well becomes a vector $e \in \mathbb{R}^{3773}$, z-scored across the field. Everything on the previous slides is one of these, and each one is a difference of means or a held-out classifier, so a lab can check it.

1 · EMBEDDING

[1]

$$e = \varphi(\text{image}); \varphi \in \{\text{CellProfiler}, \text{CNN}, \text{ViT}\}$$

Which φ barely matters: the three performed similarly on hepatocytes. The extractor is not the asset.

2 · LAB CLASSIFIER

99.9%

$$f: e \rightarrow \text{lab}, \text{ held out by plate}$$

Logistic regression, L2, five folds. Balanced accuracy, chance 1/11. Permuted labels 9.3%. Without the 1536-well labs: 100.0%.

3 · DISTANCE, THE CERTIFICATE

[e6]

$$d = \| E[e \mid \text{plate}] - E[e \mid \text{field}, c] \|_{\sigma}$$

Difference of two means in units of the field's spread, conditioned on c . Its direction is what moved; $-d$ is the way back.

4 · VARIANCE DECOMPOSITION

[e5]

$$\text{Var}(e) = V_{\text{model}} + V_{\text{lab}} + V_{\text{batch}} + V_{\text{plate}} + V_{\text{well}}$$

Nested, so shares sum to one. Microscope model separates from lab because models are shared across labs.

5 · REGRESS VARIABLES OUT

24%

$$e_{\perp} = e - \hat{E}[e \mid c], \text{ then re-run } f$$

Remove only what you know is instrument, keep what may be biology. Remove each plate's own control mean and f falls to 24%.

6 · MORPHOLOGY CHANGE, TWO WAYS

 δ

$$\text{DISTANCE } \delta = f(\|e - E[e \mid \text{vehicle}]\|), f \text{ bounded}$$

$$\text{CLASSIFIER } g: e' \rightarrow \text{label}, e' \text{ aligned to the reference before comparison}$$

Same vector, two questions. Discovery wants invariance to the lab: align first. QC wants the lab coordinate: never align it away.

results/e1, e1b, e5, e6 · [1] Ewald et al. 2026

What JUMP records. What nobody does.

Normal is a conditional expectation, $E[x \text{ / every variable below}]$. Even JUMP, built to be comparable, records only the instrument side.

RECORDED IN JUMP

- lab
- batch date
- plate
- plate type
- well position
- microscope
- confocal or widefield
- laser or LED
- objective NA
- fields per well
- filter set
- pixel size
- brightfield planes
- CellProfiler version

NOT IN ITS PUBLIC METADATA

- DMSO % per well
- DMSO lot
- DMSO water uptake
- stock freeze-thaw
- serum lot
- media lot
- passage
- thaw date
- seeding density
- confluence at dosing
- dosing to fixation
- stain and dye lots
- operator
- incubator shelf
- edge evaporation
- mycoplasma status
- CO2 and humidity

NORMALISE THE LAB AWAY AND THE LAB
DISAPPEARS

99.9%

raw features
balanced accuracy naming
the lab

24%

plate-centred
own control mean removed;
chance 9%

19%

plate-centred and scaled
own control spread
removed too

29,943 wells, 16 per plate, plates held out.
results/e1-redteam.

Centre each plate on its own controls, the standard first step, and the lab is named 24% of the time. It removes most of the coordinate a certificate needs.

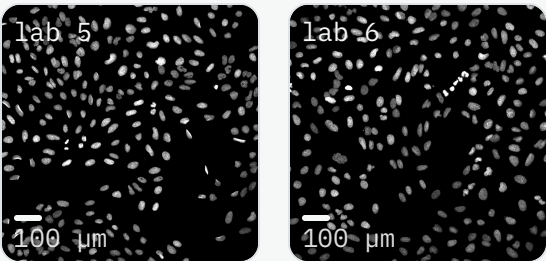
And the benchmarks are so weak that **counting cells** often wins.

If a naive baseline matches your model, the benchmark is the problem. The bottleneck was never the model. It is what you are comparing against.

Two ways to ask the same question.

Why now: an image classifier can answer this, and until recently nothing could. A count moves only when cells die or divide; an embedding of the whole well moves when anything changes.

SAME CELL LINE, SAME PROTOCOL



cell count	236	224	same
shape	0.94	0.50	different

METHOD ONE

Does this run belong with the past runs?

$$g: e' \rightarrow \text{belongs} / \text{does not}, \quad e' = \text{align}(e)$$

Align first, so the answer is the cells and not the day.

Standard alignment; the tuning is ours.

METHOD TWO

How far out, and which way back?

$$\delta = f(\|e - E[e \mid \text{vehicle}]\|), \quad f \text{ bounded}$$

Bounded, so one wild well cannot swamp the score.

Distance published with the Broad [1]; transform unpublished.

Both methods need the same thing to work: somebody else's images. It takes a reference to know which.

Fig. A5 Same cell line, two untreated wells, and the two readouts. Left, DNA channel, JUMP cpg0016, CC0; nuclei by Otsu segmentation; detail = high-frequency energy over total, inside each nucleus. Right, the two readouts: a classifier on aligned embeddings, and a bounded distance from the vehicle centre. Alignment and distance follow published practice [1]; the specific transform and tuning are not disclosed here. [1] Ewald JD, Titterton KL, et al. *Cell Systems*, 2026.

The line is not expired. Your baseline is.

Two plates of one lab sit further apart the longer the gap between them. In lab 2 the distance is still rising after thirteen weeks.



IS THE CELL LINE RUINED?

Not known from profiles alone. The data says the lab moved and when. It does not say why.

SO WHY RE-CERTIFY?

Because your certificate is a comparison made on one day, and the baseline it was made against goes stale within weeks.

HOW SOLID IS THE NUMBER?

One lab, 219 plates, 13 batches, June to October 2021. 0.63 in the same batch, 4.7 over 13 weeks. Other labs are not yet measured.

Median distance between plate centroids, in within-plate spreads, with the interquartile range. results/e1b-drift.

STR says who.

Images say how.

Not a competitor: the other half of the same certificate. Every lab already runs the left column. The right column costs nothing extra.

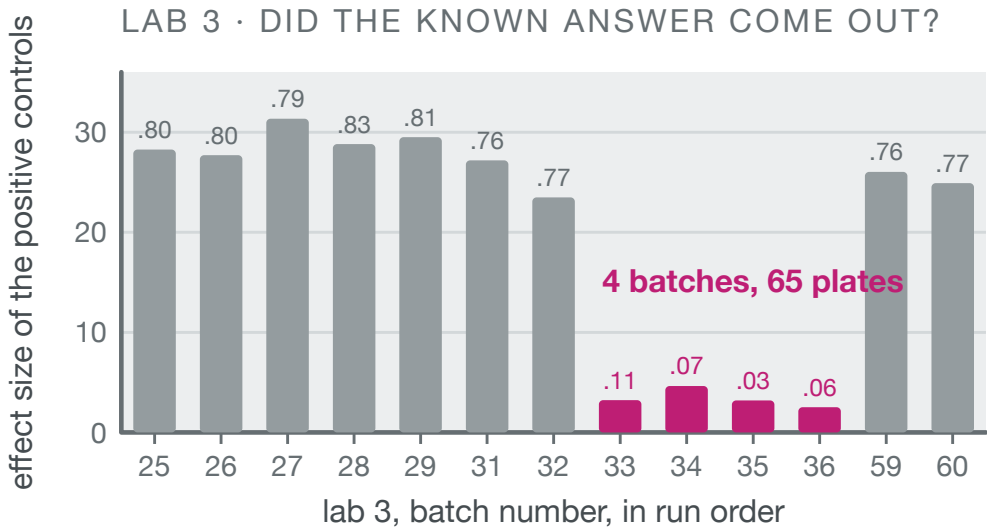
	STR PROFILING	CONTROL-WELL IMAGES
WHAT IT READS	DNA at 17 short tandem repeat loci	the pictures the screen already took
WHAT IT PROVES	the line is what the label says	the cells still behave like the field's version of that line
WHAT IT MISSES	two cultures, same fingerprint, different behaviour	identity: it cannot tell you the line is mislabelled
SAMPLE	cells spotted on a card and mailed	nothing leaves the building but images
EXTRA WORK	a separate prep, per line, per test	none: control wells you already image
CADENCE	on receipt, after 10 passages, when in doubt	every quarter, from the last plate you ran
ANSWER	a profile to compare by eye or by database	one of two words, and a distance

Identity once. Behaviour every plate. One line on the certificate each.

STR practice and cadence: ATCC cell authentication service and test recommendations; 17 loci plus amelogenin on the Promega PowerPlex 18D system. Low- and high-passage LNCaP respond differently to the same hormones: ATCC technical document.

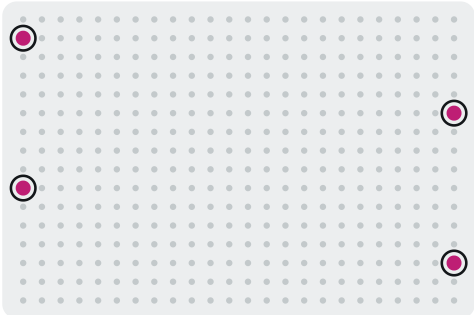
A known answer, run in every lab.

Untreated wells certify where a lab sits. Positive controls certify that the assay answers at all. The field already runs the same eight.



WHERE ON THE PLATE DOES AMG900 SHOW?

a batch that came out
plate C13443aW



○ where the plate map puts it

a batch that failed
plate A12069aW



● wells that look like it

RUNS NOW ON JUMP

Every compound plate carries eight positive controls. In 125 of 129 batches the known answer came out.

WHAT THE FIELD CAUGHT

Lab 3, 4 consecutive batches, 65 plates: the wells the plate map labels as positive controls show no effect, and no other well on the plate does.

TWO CHECKS, TWO FAILURES

The untreated-well certificate flagged 1 of these 4 batches. A plate can sit in the right place and still not answer.

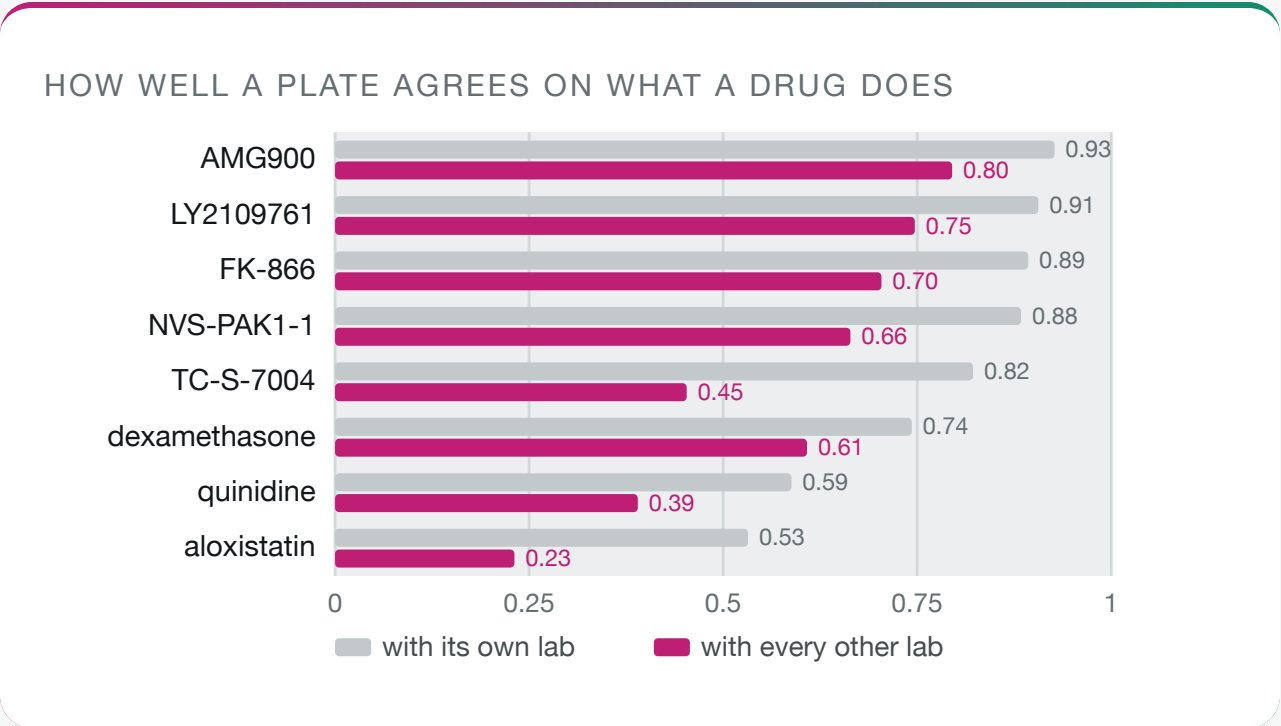
Same drug. Different answer.

The field already contains the experiment: eight fixed compounds, each measured 1,829 times, in 11 labs.

WHAT A SCREEN REPORTS, PER PLATE

$$\delta_{p,c} = E[x \mid \text{plate } p, \text{ compound } c] - E[x \mid \text{plate } p, \text{ DMSO}]$$

Treated minus control, on the same plate. Scaling fit on the field's DMSO wells only, never on the plate being judged.



Q1 · DOES IT AGREE? [E8]

Less with other labs, for every compound.
AMG900: 0.93 with its own lab, 0.80 with the others.

Q2 · DOES IT TRACK THE CONTROLS? [E8]

Inside labs, yes.
Spearman 0.20 to 0.53 by compound; pooled 0.30, shuffled 0.01. Between labs: same sign, too few labs.

Q3 · DOES IT MATTER? [E8]

Effect size differs 14% to 79% between labs.
Inside one lab: 10% to 44%.

Shown in fluorescence.

Bet on brightfield.

Every number here is Cell Painting: five stains. The product reads brightfield: none. That gap is the bet, testable on public data.

WHAT IS SHOWN

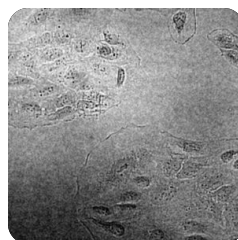
Stained wells name the lab

Untreated U2OS wells, 11 labs: the lab is named 99.9% of the time, chance 9%.
results/e1

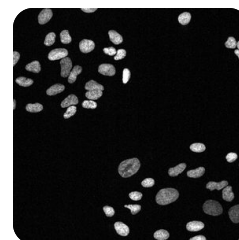
WHY THE BET IS FAIR

Fewer variables, not more

No stain lot, dye lot or staining step:
three unrecorded variables gone (A3).
JUMP images brightfield beside the
stains, so the same wells re-run without
them.



BRIGHTFIELD



DNA STAIN

One field of
one well,
two ways.
Lab 2, June
2021.

WHAT WOULD KILL IT

Brightfield alone at chance

Then the certificate asks for one cheap
stain. The test: e1 on the brightfield
planes of the same 93,228 wells.
[pending]

Brightfield timecourse is the modality I build screens on.

Would it cry wolf? 6 of 129.

A flag is only worth something if it is rare, and if it survives the choices made to compute it. Both are measured.

LAB 2 · THE SAME 13 BATCHES, JUDGED TEN WAYS

	7	14	21	12	19	26	2	8	16	23	30	20	3	
	Jun	Jun	Jun	Jul	Jul	Jul	Aug	Aug	Aug	Aug	Aug	Sep	Oct	
as on slide 06	■	■	■	■	■	■	■	■	■	■	■	■	■	3,773 features
baseline is the first two batches	■	■	■	■	■	■	■	■	■	■	■	■	■	3,773 features
baseline is the first three batches	■	■	■	■	■	■	■	■	■	■	■	■	■	3,773 features
no clipping	■	■	■	■	■	■	■	■	■	■	■	■	■	3,773 features
no image-level features	■	■	■	■	■	■	■	■	■	■	■	■	■	2,904 features
image-level features only	■	■	■	■	■	■	■	■	■	■	■	■	■	869 features
all-vehicle plates removed	■	■	■	■	■	■	■	■	■	■	■	■	■	3,773 features
lab 1 removed from the field	■	■	■	■	■	■	■	■	■	■	■	■	■	3,773 features
lab 8 removed from the field	■	■	■	■	■	■	■	■	■	■	■	■	■	3,773 features
labs 1 and 8 both removed	■	■	■	■	■	■	■	■	■	■	■	■	■	3,773 features

HOW RARE

6 of 129 batches are outside, in 3 of 11 labs. 8 labs never flag.

DOES EVERY PLATE AGREE

In lab 2's flagged batches 25 of 25 plates are outside. In its other 10 batches, 2 of 194.

WHAT CHANGES IT

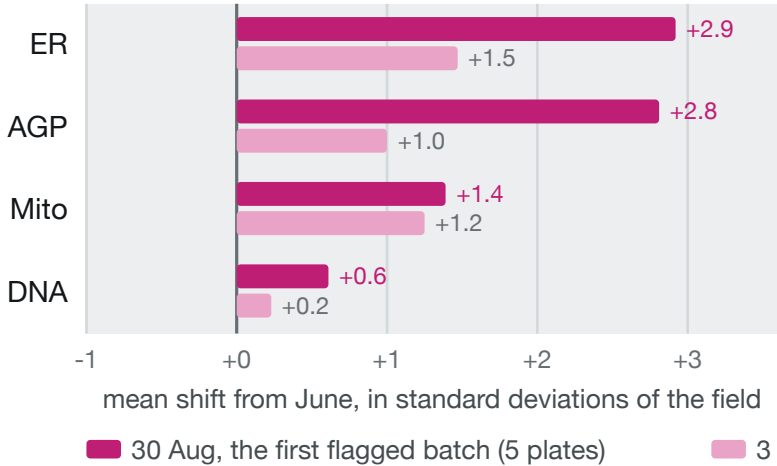
1 of 10 variants: a baseline that takes in 21 June, the batch whose cell count fell. A baseline should be a batch the lab trusts.

Outside: further from its own first batch than from the nearest other lab. Flagged batches hold 5, 6, 14 plates.

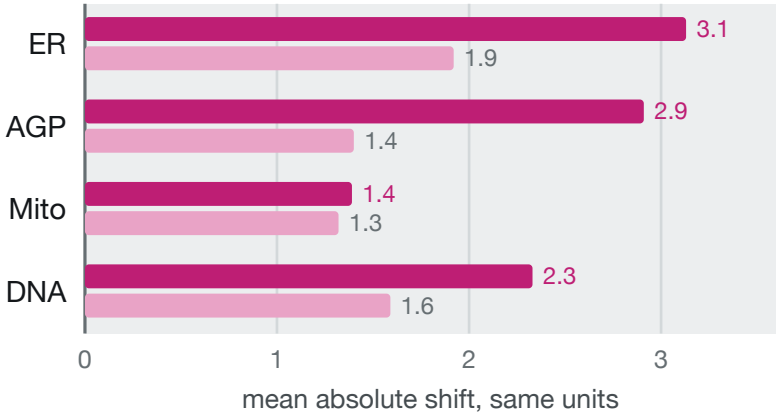
Stain intensity rose. Cell count did not.

The certificate's distance is a difference of two means, so it reads feature by feature.

STAIN INTENSITY



TEXTURE



READING

On 30 Aug, ER and AGP intensity sit 2.9 and 2.8 sd above June, and 96% or more of those features moved the same way. By 3 Oct: 1.5 and 1.0. Cells per well: 189, then 165 and 182.

INSTRUMENT OR CELLS

Image-level features carry 20% of the move and are 23% of the features, so the move is not confined to acquisition. Profiles cannot tell a stain lot from biology.

WHAT A LAB GETS

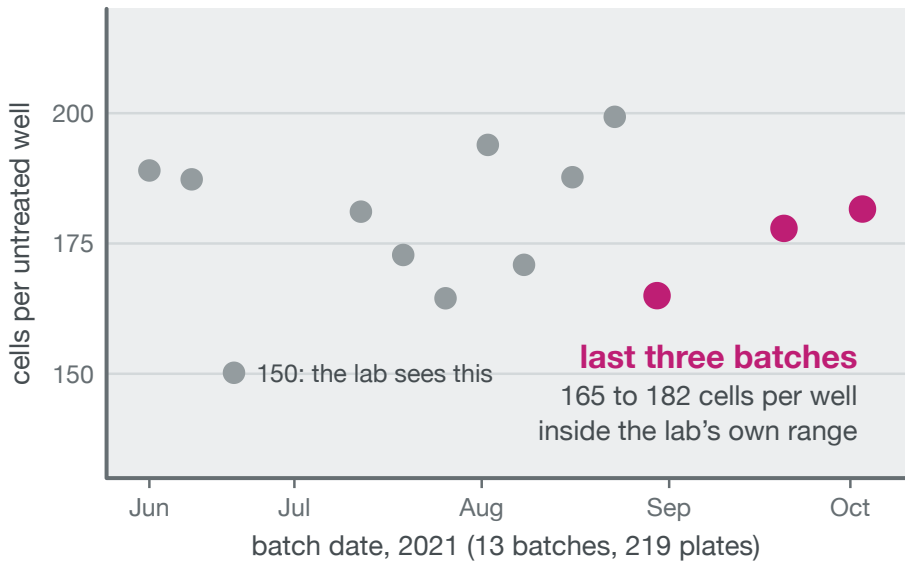
Where to look first: staining and exposure in two channels. Not why. The cause is the lab's to find, a day after the run and not four months.

Shift from the lab's first batch, in standard deviations of the field; means over the features of one channel.

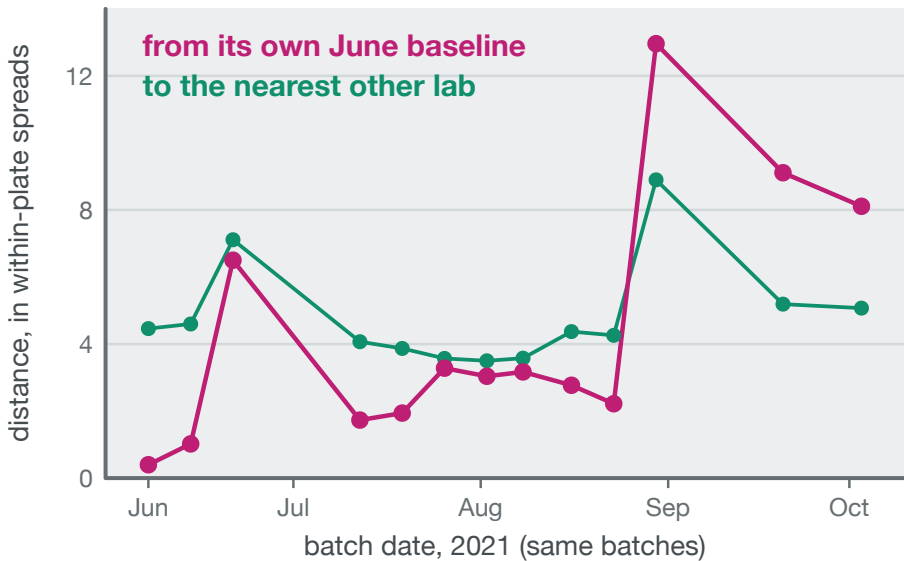
One lab, 13 batches.

The lab’s own check and the field’s view of the same batches, side by side.

WHAT THE LAB CAN SEE



WHAT ONLY EVERYONE ELSE CAN SEE



THE LAB’S OWN CHECK

Cells per untreated well stay between 150 and 199. The last 3 batches sit inside the range of the earlier ones.

THE FIELD’S VIEW

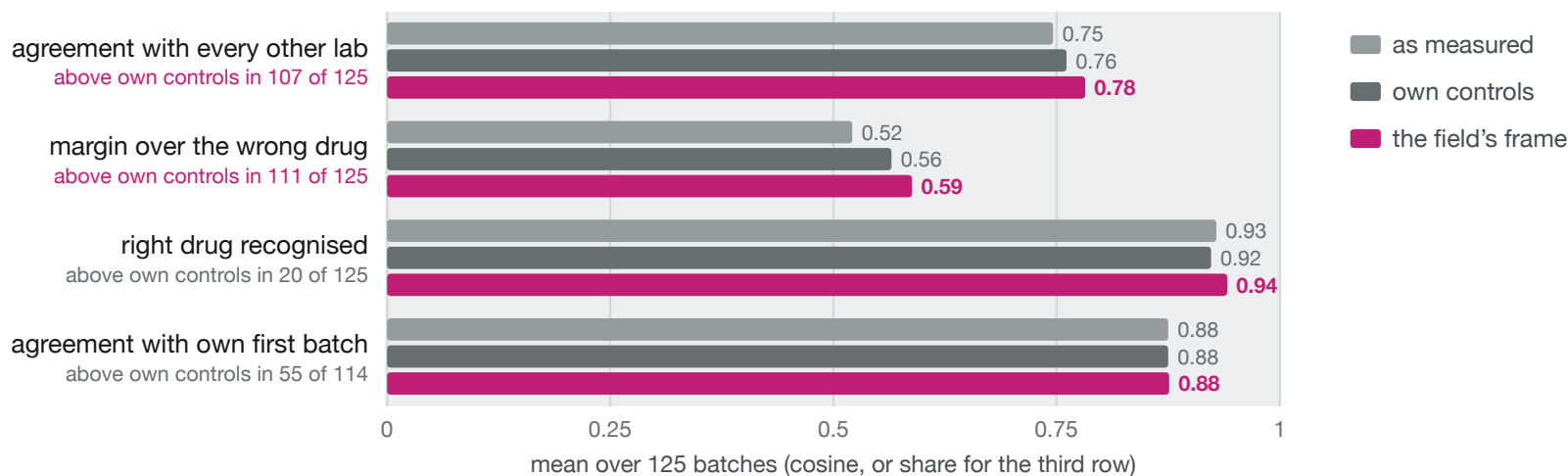
From 30 Aug the lab is further from its own June baseline than from another lab. 3 Oct: 8.1 from baseline, 5.1 to lab 8.

WHY BOTH

Neither replaces the other. The count is fast and local. The field’s view needs everyone else’s images.

One frame: **what it buys.**

Each batch's untreated wells are moved onto the field's, built without that lab. The drugs are the test.



DOES IT HOLD

With 100, 200 or 400 components and shrinkage from 0.05 to 0.5: above own controls in 89 to 116 of 125 batches for agreement, 52 to 123 for margin.

WHAT IT DOES NOT DO

A lab's agreement with its own first batch does not change: 0.88, then 0.88. Effect sizes vary more between batches: 0.06 becomes 0.12. The gain is between labs, and it is modest: 0.76 to 0.78 over own controls.

HOW DENSE

A lab's own baseline settles by eight plates: 0.89 to 0.39 spreads. The field's reference moves 0.41 when one lab leaves.